

PKCERT AI & Software Development Internship

Task 06: Exploratory Data Analysis with a Jupyter Notebook

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Overview

This report is the written companion to my Task 06 Jupyter notebook, `EDA_Wine_Quality.ipynb`. The task was to take a real dataset, explore it properly, and write up what the data is actually telling us. The notebook does the hands-on work with markdown notes at every step, and this report pulls the story together with the same figures and the key takeaways.

I picked the **Red Wine Quality** dataset from the UCI Machine Learning Repository. It is a nice fit for an EDA exercise: every column is numeric, it has a clear target to explain, and it has just enough data-quality quirks to be realistic without being a mess.

Part A: The Dataset

- **Source:** UCI Machine Learning Repository, originally from Cortez et al. (2009).
- **Purpose:** Each row is a Portuguese red *vinho verde* wine. Eleven physicochemical lab measurements were recorded, and a panel of tasters scored each wine. The question is whether the chemistry can explain the score.
- **Records:** 1,599 wines to start with.
- **Features (11, all numeric):** fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, and alcohol.
- **Target:** quality, an integer taster score that runs from 3 to 8 in this data.

Part B: Exploratory Data Analysis

Structure and Summary

The table is 1,599 rows by 12 columns, and every column comes in as a number, so there is nothing to decode or convert. The summary statistics show the columns live on very different scales: total sulfur dioxide runs into the hundreds while density sits right around 1.0. The quality score averages about 5.6, so most wines are middling rather than brilliant or awful.

Data Quality

The dataset is unusually tidy on missing values: there are none at all. It does have **240 fully duplicated rows** though. Those might be genuinely different wines that recorded identical measurements, but to keep them from getting double weight I dropped them, which leaves **1,359 unique rows**. There were no other inconsistencies to fix: every measurement is positive and sits in a physically sensible range, for example pH between roughly 2.7 and 4.0, which is exactly where wine should be.

Visualizations

The notebook produces six figures, each with a short interpretation. They are reproduced here.

1. Distribution of the quality score

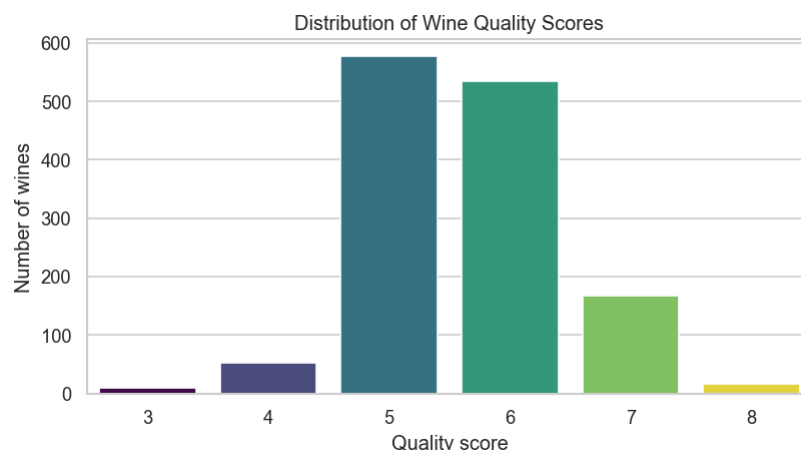


Figure 1: How many wines sit at each quality score.

The scores pile up on 5 and 6, with very few wines at 3, 4, or 8. This imbalance is worth remembering: there simply are not many examples of great or poor wines to learn from.

2. Distribution of each chemical feature

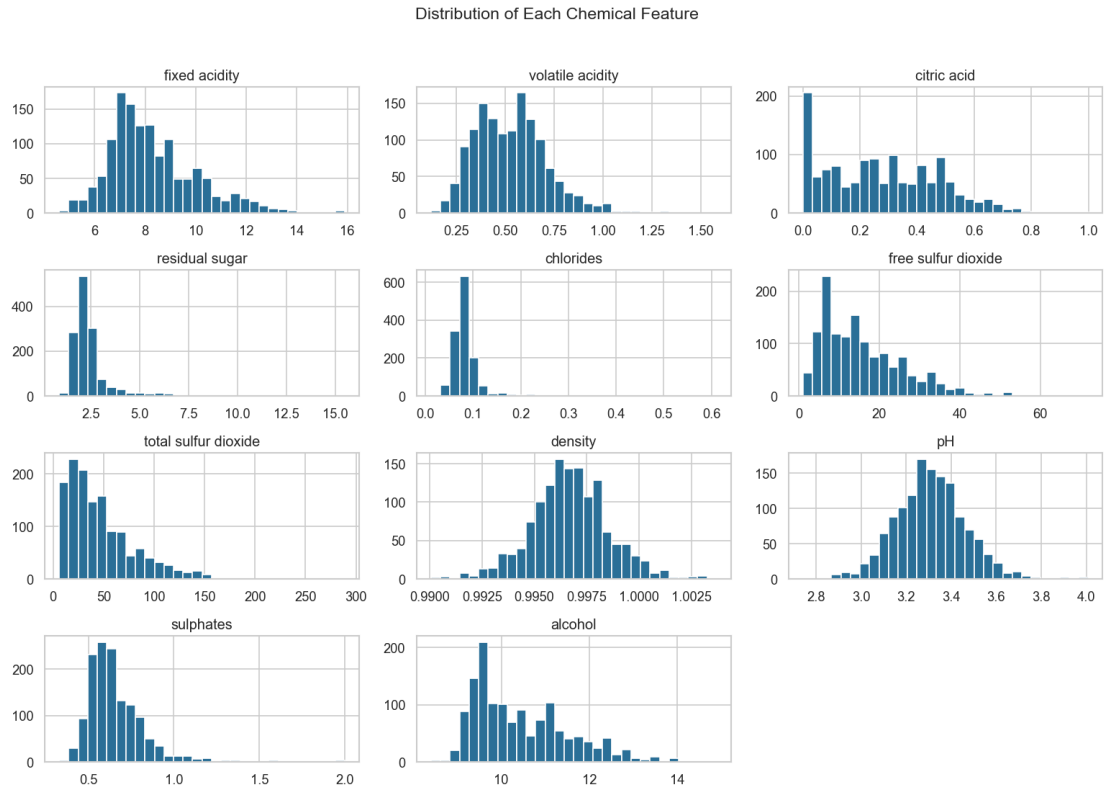


Figure 2: Histograms of all eleven features.

Most features are right skewed with a long tail of high values. Residual sugar and chlorides are the most extreme, with a few wines sitting far out to the right. That skew is a hint that a handful of wines are chemically unusual and that scaling would help before any modelling.

3. Correlation heatmap

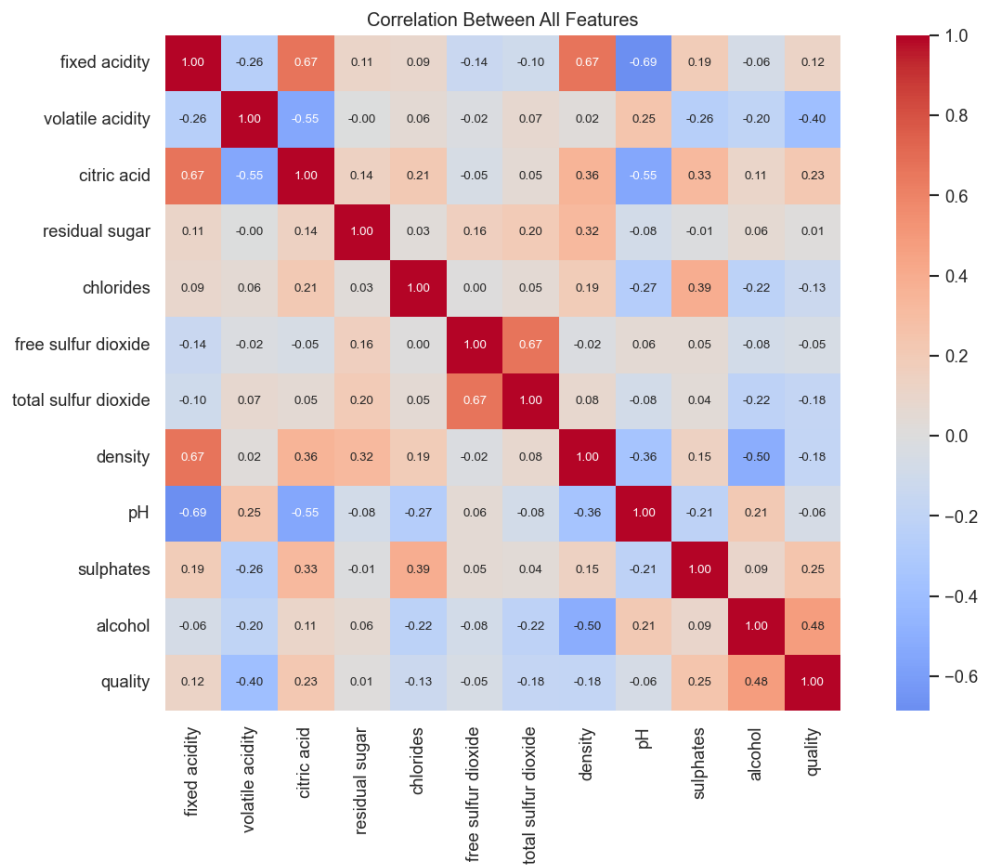


Figure 3: Correlation between every pair of features.

This is the most useful single view. Reading the quality row, alcohol has the strongest positive correlation (about +0.48) and volatile acidity the strongest negative one (about -0.40). Among the features themselves, fixed acidity ties to citric acid and density, and the two sulfur dioxide columns move together, all of which is chemically sensible.

4. Alcohol by quality

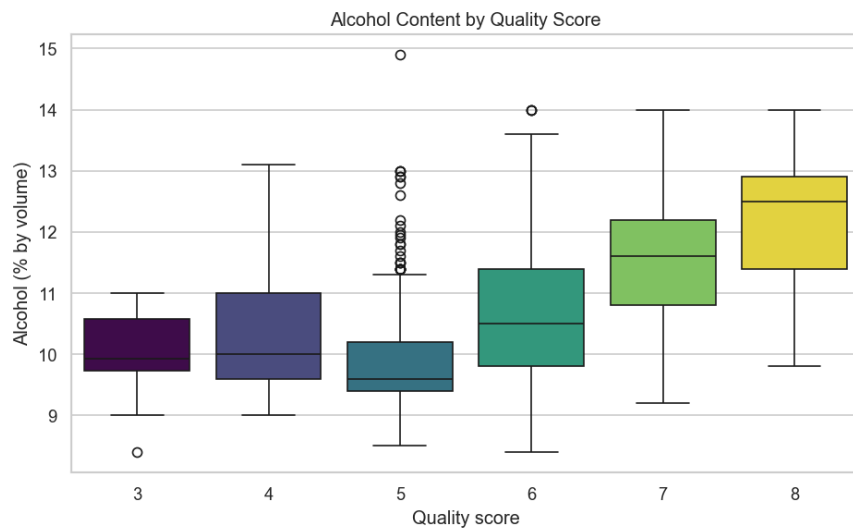


Figure 4: Alcohol content for each quality score.

This is the clearest relationship in the data. From score 5 upward, the median alcohol content climbs at every step, so better rated wines tend to be more alcoholic. It is the same signal the heatmap flagged, now easy to see.

5. Volatile acidity by quality

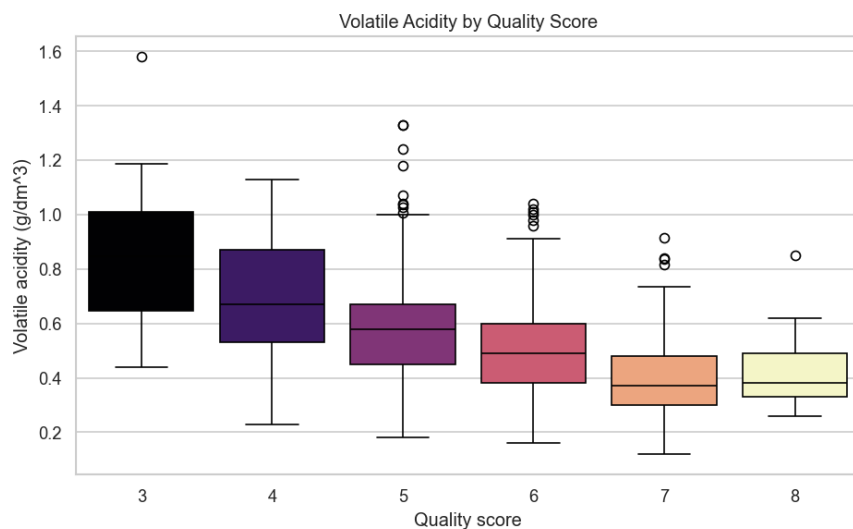


Figure 5: Volatile acidity for each quality score.

This chart is the mirror image of the last one. Volatile acidity is what gives wine a sharp vinegar note, and its median drops as quality rises. Tasters clearly marked down the more acidic wines.

6. Correlation of each feature with quality

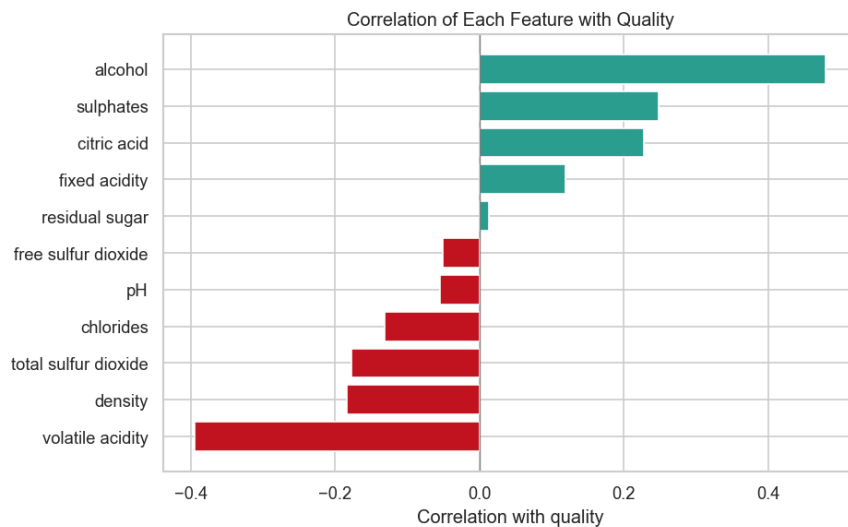


Figure 6: Every feature's correlation with the quality score.

Lining the numbers up in one bar chart gives a tidy summary. Alcohol, sulphates, and citric acid help a wine's score, while volatile acidity hurts it the most, followed by total sulfur dioxide, density, and chlorides. Everything else sits close to zero.

Part C: Findings and Insights

Key Findings

- Quality rides on just a few features: alcohol (positive) and volatile acidity (negative) are by far the two biggest signals, with sulphates and citric acid adding a smaller positive push.
- Higher alcohol goes hand in hand with higher ratings.
- Sharp, acidic wines score badly.
- The target is imbalanced, with most wines rated 5 or 6.

What This Could Mean

For a winemaker, the practical read is that aiming for a strong fermentation (higher alcohol) while keeping volatile acidity low is the chemical combination most tied to a better score. Sulphates, which work as a preservative, also line up with slightly better ratings. From a research angle, no single feature correlates above about 0.5 with quality, which says the rating is genuinely a multi-factor thing. That points towards a model that combines features rather than leaning on any one measurement.

Limitations and Next Steps

The biggest limitation is the class imbalance: with so few wines at the extremes, it is hard to say much about what makes a wine truly great or truly poor. The target is also subjective, since quality is an average of human taste scores. And because these are all red *vinho verde* wines from one region, the findings may not carry to other wines. A natural next step would be to fold in the white wine dataset for a bigger and more balanced picture, group the scores

into low, medium, and high to ease the imbalance, and then train a classifier to see how well the chemistry actually predicts quality on unseen wines.

Part D: Notebook Organization

The notebook is laid out to follow this same structure. It opens with the dataset description, moves through the exploratory analysis (structure, data quality, then the visualizations with a note under each), and closes with the written insights. Markdown headings separate every stage, the code cells are commented, and each figure is explained right where it appears. All the charts are saved into the **figures/** folder, which is how they made their way into this report.

The full project, including the notebook, the dataset, the figures, and this report, is on my GitHub repository for the internship: <https://github.com/AbdullahAmir06/ai-internship-abdullahamir>.